

MS21250 — 12-Point External Wrenching Tension Bolt

High-strength alloy-steel aerospace bolt · 180 ksi tensile / 108 ksi shear · 450°F

Designations & cross-reference

CALLOUT	ROLE	STATUS
MS21250	Military Standard part sheet (dimensions, head form)	Active; redesignated NASM21250
NASM21250 (rev L, 2013)	Current AIA/NAS part standard	Current designation
MIL-B-8831B (1982)	Procurement spec (material, strength, acceptance)	Cancelled 3 Apr 2000 → NASM8831
NASM8831	Successor procurement specification	Current material spec
NAS624–NAS644	Related 12-point external-wrenching bolt family	Adjacent standards

Mechanical properties

Ultimate tensile (Ftu)	180,000 psi (180 ksi) min
Ultimate shear (Fsu)	108,000 psi (108 ksi) — 0.60 × Ftu
Hardness	39–43 HRC
Service temperature	450°F (finish-limited)
Head	12-point external wrenching, flanged
Thread	UNJF-3A, right-hand, controlled root

Material & finish

Alloy family	4140, 4340 / E4340, 6150, 8735, 8740 (aircraft-quality low-alloy steel)
Finish	Cadmium plate + chromate per (AMS-)QQ-P-416 Type II, Class 2
Post-plate	Hydrogen-embrittlement relief bake (AMS2759/9)

Why 450°F? The ceiling is set by the cadmium plating, not the steel: cadmium embrittles high-strength steel as it nears its ~600°F melt point. At 180 ksi / 39–43 HRC the bolt sits within QQ-P-416's cadmium limits and is fully conforming.

Part-number decoder — MS21250[H]-DD-LLL

FIELD	MEANING	EXAMPLE
H (optional)	Drilled head for safety wire (omit = undrilled)	MS21250H04004
DD	Diameter in 1/16" units	04 = 1/4", 05 = 5/16"
LLL	Grip length in 1/16" units	008 = 0.500"

Two details most listings get wrong: (1) Dash 03 = 0.190" (#10), not 3/16". (2) Odd grip dashes carry a +0.061" offset, so code 022 ≈ 1.375"; the clean divide-by-16 rule is exact only for even codes.

Diameter dash codes & grip-coverage matrix

DASH	NOMINAL DIA.	COMMONLY STOCKED GRIP DASH CODES
03	0.190" (#10)	04, 06, 08, 10, 12, 16
04	1/4"	04, 06, 08, 10, 12, 13, 14, 16, 18, 20, 22, 24, 26, 28, 30, 34, 38, 40, 42, 46, 52, 54
05	5/16"	06, 08, 09, 10, 12, 14, 15, 16, 17, 18, 22, 24, 26, 28, 30, 32, 36, 40, 48, 56, 62
06	3/8"	06, 08, 10, 12, 14, 16, 18, 20, 22, 24, 26, 30, 34, 38, 40, 42, 44, 56, 62
07	7/16"	04, 08, 14, 16, 20, 22, 30, 64
08	1/2"	18, 20, 24, 26, 27, 28, 31, 32, 34, 38, 42, 47, 50, 62, 76
09	9/16"	14, 18, 20, 22, 24, 38, 66
10	5/8"	22, 24, 70
12	3/4"	24, 36, 40

14	7/8"	46
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Grip dash → inches: divide by 16 (e.g. 022 = 1.375"). Diameter callouts for the smallest codes map to the nearest standard screw size; confirm against NASM21250 for controlled procurement. Per-size minimum shear/tensile loads in pounds are taken from the controlling drawing.

Supply & quality

TorqBolt manufactures bolting to MS21250 / NASM21250 requirements from works in Mumbai and Rajkot, India, under an **ISO 9001** quality system, with **PED 2014/68/EU** and **AD 2000-Merkblatt** material capability and full heat-to-bolt traceability. RFQ: info@torqbolt.com · **+91-22-66157017** · ms21250.com

Data sources: Cross-checked against the U.S. DLA/FLIS characteristics record, the NASM21250 / NASM8831 standards, and federal spec QQ-P-416. This datasheet is a reference summary; confirm all values against the controlling NASM21250 document before design or procurement. © TorqBolt. Not an aerospace-prime-approved release; ISO 9001 manufacturing to the MS21250/NASM21250 requirement set.